

NOWRA, Australia

WMO: 947500

Lat: 34.9469S

Lon: 150.5353E

Elev: 344

StdP: 14.51

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.7	41.4	27.1	21.4	61.6	29.8	24.2	57.4	30.1	55.1	27.8	55.4	6.5	290	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	92.0	68.2	87.5	68.2	83.9	67.6	73.5	82.9	72.0	80.5	70.7	78.7	13.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	116.1	76.1	69.7	110.5	74.7	68.3	105.2	73.6	37.4	83.0	35.9	80.7	34.8	78.8	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	21.9	19.0	DB	35.4	103.0	1.9	4.6	34.0	106.3	32.9	108.9	31.9	111.5	30.5	114.8	(4)
(5)			WB	34.0	76.5	1.9	1.9	32.7	77.9	31.5	79.0	30.5	80.1	29.1	81.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	70.6	70.0	67.7	62.9	57.9	53.9	52.5	54.0	58.0	61.6	64.7	68.1	(6)	
	DBStd	7.81	5.39	4.85	4.38	4.27	3.84	3.16	3.30	4.06	5.06	5.70	5.95	5.38	(7)	
	HDD50	26	0	0	0	0	1	4	12	8	1	0	0	0	(8)	
	HDD65	1871	9	6	21	90	222	333	389	341	218	139	77	26	(9)	
	CDD50	4326	638	560	549	386	246	122	88	133	242	360	440	562	(10)	
	CDD65	696	182	147	105	26	2	0	0	1	9	34	67	123	(11)	
	CDH74	5352	1344	948	581	181	17	0	1	13	153	440	639	1035	(12)	
	CDH80	1861	526	322	154	38	0	0	0	1	36	154	242	388	(13)	
(14) Wind	WSAvg	8.6	7.9	7.6	7.4	7.4	8.7	9.3	10.1	10.4	9.4	8.7	8.3	8.2	(14)	
(15) Precipitation	PrecAvg	42.3	3.9	4.5	5.0	4.2	3.5	4.4	2.0	2.7	2.8	3.6	3.9	2.8	(15)	
	PrecMax	75.9	13.1	17.7	16.9	19.8	12.9	24.0	6.1	14.4	9.3	16.2	23.0	8.9	(16)	
	PrecMin	18.2	0.4	0.1	0.1	0.0	0.1	0.3	0.0	0.0	0.2	0.2	0.1	0.1	(17)	
	PrecStd	14.4	2.8	4.3	4.6	4.4	3.0	4.7	1.7	3.5	2.4	3.3	4.1	2.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.0	95.9	89.2	85.2	76.4	69.6	69.8	75.5	84.6	91.6	95.2	97.2	(19)	
		MCWB	70.0	69.5	68.0	62.1	59.8	56.3	53.9	57.4	59.5	62.2	64.6	68.0	(20)	
	2%	DB	89.9	88.1	84.0	78.7	72.3	65.7	65.5	70.2	78.2	84.1	86.6	88.8	(21)	
		MCWB	70.6	72.2	69.0	63.3	59.0	55.1	52.9	54.7	58.0	61.5	65.8	66.4	(22)	
	5%	DB	85.7	83.3	80.1	74.8	69.0	63.3	62.6	66.5	73.5	78.6	80.7	83.6	(23)	
		MCWB	70.7	70.4	67.9	62.4	57.9	53.7	51.6	53.1	57.1	60.8	65.7	67.1	(24)	
	10%	DB	81.1	79.5	76.5	71.7	66.2	61.0	60.2	63.3	69.1	73.8	76.2	79.5	(25)	
		MCWB	69.1	69.0	66.3	61.5	56.9	52.6	50.7	51.9	56.0	59.9	63.9	66.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.4	76.0	74.0	68.3	65.0	60.1	58.2	60.4	64.4	68.1	71.8	73.4	(27)	
		MCDB	87.0	85.7	81.4	76.9	69.2	64.9	63.5	68.6	73.7	78.2	83.0	84.3	(28)	
	2%	WB	73.2	73.7	71.7	66.2	62.6	57.5	55.7	57.4	61.8	65.6	69.0	71.2	(29)	
		MCDB	83.3	82.5	78.1	73.4	66.5	61.6	60.4	64.6	71.0	75.2	79.8	81.3	(30)	
	5%	WB	71.7	71.9	69.9	64.6	60.4	56.0	54.0	55.5	59.8	63.7	67.2	69.6	(31)	
		MCDB	80.9	79.3	76.3	70.9	65.6	60.0	58.7	62.3	68.5	72.9	76.7	79.0	(32)	
	10%	WB	70.3	70.4	68.2	63.1	58.5	54.6	52.3	53.7	57.7	61.8	65.3	68.1	(33)	
		MCDB	78.4	76.9	73.8	69.2	64.4	58.7	57.7	60.5	65.9	70.5	73.5	75.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	16.1	16.2	17.1	16.6	14.9	16.2	18.4	19.7	20.0	18.5	18.4	(35)	
		MCDBR	27.2	24.4	23.5	22.7	20.8	18.4	20.2	23.6	28.1	30.0	28.7	28.0	(36)	
	5% WB	MCWBR	11.0	10.2	10.8	10.5	10.4	9.9	10.5	11.4	11.9	12.3	11.6	10.5	(37)	
		MCWBR	22.3	20.0	19.3	19.2	17.2	14.8	16.6	20.1	23.4	23.7	23.9	22.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.396	0.378	0.354	0.326	0.309	0.299	0.290	0.300	0.331	0.348	0.379	0.381	(40)	
		taud	2.422	2.503	2.549	2.603	2.598	2.629	2.642	2.604	2.515	2.494	2.441	2.452	(41)	
	Ebn at Noon		298	297	293	284	272	267	277	289	296	303	302	304	(42)	
		Edh at Noon	39	35	31	26	23	21	22	25	31	35	38	38	(43)	
(44) All-Sky Solar Radiation	RadAvg		2015	1715	1475	1187	943	734	867	1143	1536	1798	1893	2064	(44)	
	RadStd		164	151	118	102	55	63	69	68	96	146	162	165	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.77	+0.79	N/A	N/A	N/A	N/A	N/A	-205	+347	+167	(46)	
(47) Regional (32 neighbors)		+0.79	N/A	N/A	+1.37	N/A	N/A	-2	-153	+328	+167	(47)	

Nomenclature: See separate page